Semantic Simultaneous Location and Mapping on Embedded Systems

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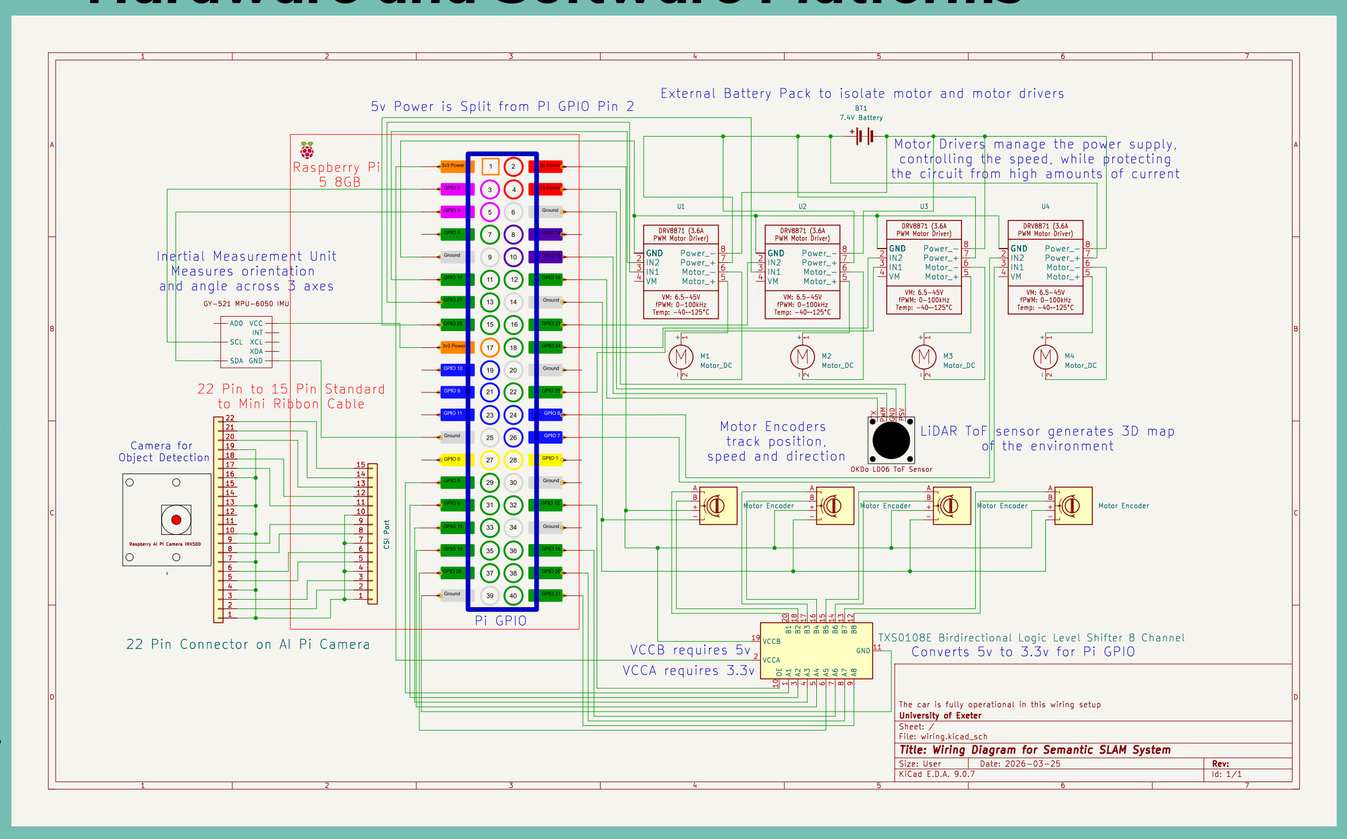
Introduction

- Modern SLAM has evolved into semantic labelling of an environment [9].
- Most current Semantic SLAM algorithms depends on large amounts of computing power to achieve an accurate map.
- This project aims to create a lightweight mobile semantic SLAM system capable of creating a labelled occupancy map on a Raspberry Pi in real-time.
- This project will fully deliver a mobile system capable of generating accurate, real-time, labeled occupancy maps, without relying on cloud processing.

Hardware and Software Platforms

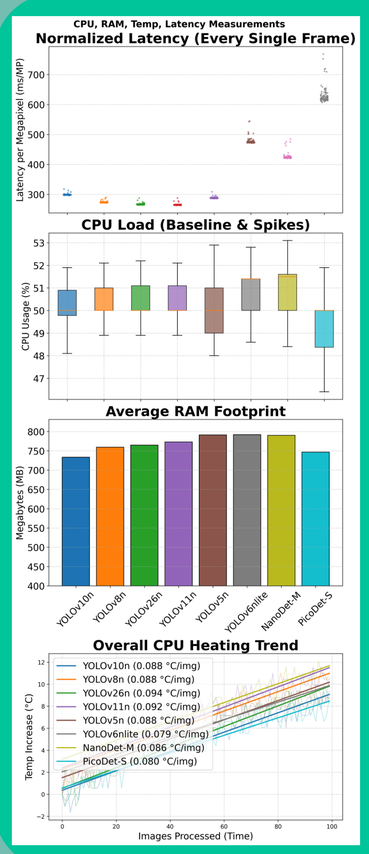
Hardware
- Raspberry Pi 5 8GB
- LD06 ToF Sensor
- IMU MPU6050 IMU Sensor
- IMX500 Pi Camera

Software
- ROS2 running on Ubuntu Noble
- YOLOv26n Object Detection Model
- All written code to handle sensor data is in C++



Object Detection

- The object detection model was chosen from testing many different architectures and models that are trained on the COCO dataset.
- The models were converted to an ONNX runtime model to improve the efficiency of all the models, also simulating the deployment of the model.
- Models that benefit from quantisation have also been adjusted to accept 8-bit integer inputs.
- The models were measured on their CPU usage, RAM usage, inference latency, and temperature



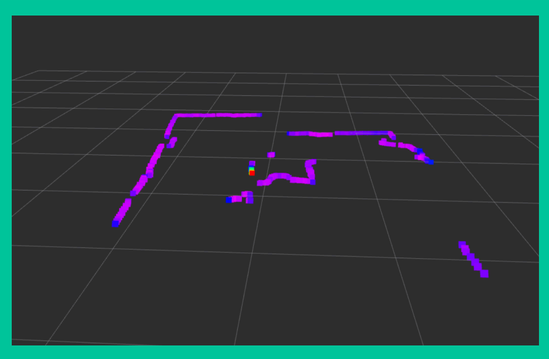
Models	mAPval 50-95	Input Size
YOLO10n[3]	38.5	640x640
YOLO8n[3]	37.3	640x640
YOLO11n[3]	39.5	640x640
YOLO26n[3]	**40.9**	**640x640**
YOLO6n[3]	37.5	640x640
YOLO5n[3]	34.3	640x640
NanoDet[1]	30.4	416x416
PicoDet-S[2]	27.1	320x320

YOLOv26n was selected for final deployment. It achieved the lowest latency, stable thermals, and the highest mAP score. Despite higher resource demands, its larger input size significantly improves object detection by preserving finer details.

Geometric SLAM

The IMU returns a series of raw integer data that represents the acceleration and the change in rotation.
The motor wheel encoders return the position, speed, and direction of the motor shaft.

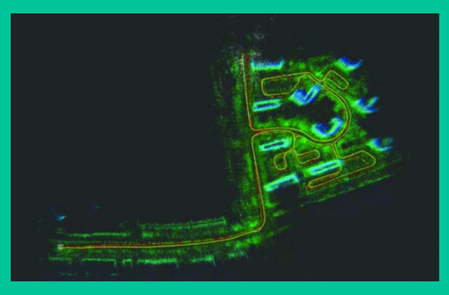
The IMU and the wheel odometry data are combined to improve localisation as it mitigates the drift of the wheel odometry and the noise from the IMU [4].

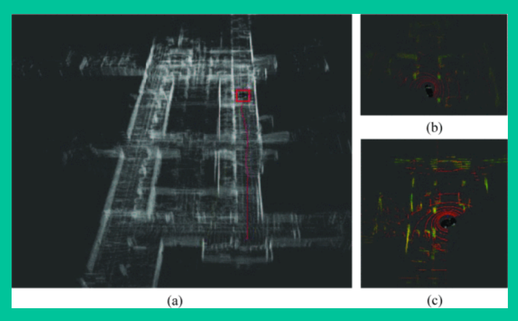


Point Cloud Data Visualisation

The LD06 ToF sensor outputs a 2D point cloud of data with the angle, distance, and confidence.
It can reach up to 12m, with a frequency of 5-15Hz

Global map from LiDAR, IMU and Wheel Odometry data by Huang et al [10]





The observation data(b-c) and trajectory(a) of the car obtained by the LiDAR, IMU, Wheel Odometry system by Zhang et al [9]

Semantic SLAM

There are many limitations when combining these two algorithms:
- Geometric SLAM algorithms often struggle with loop closure, which is when a car does not know if they have completed the entire environment [4].
- Object Detection models can often detect the wrong object, especially if the object class was not present while training [8].
- Both of these algorithms can require a lot of computational power meaning less accurate, lightweight models and algorithms must be chosen
- The geometric SLAM algorithm may need set limitations on reading sensor input data, which will limit the accuracy and confidence of the readings it recieves.

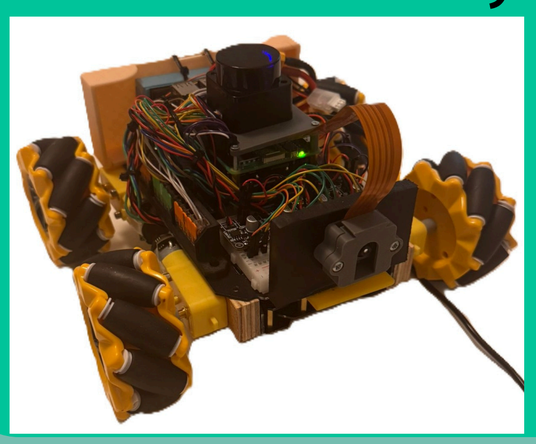
Semantic Map Example

This is a semantic map labelled by a semantic labelling program with the geometric SLAM making the underlying map [11].

A semantic map that moves beyond obstacle avoidance by overlaying object classifications directly onto the 2D occupancy map made by the geometric SLAM algorithms

Fusing the coordinates from the geometric mapping and the semantics from the object detection model can be quite challenging.
- The system dynamically fuses the geometric mapping layer and the object detection layer so that as the algorithm creates a node, a label from the neural network is assigned to it.
- The algorithm needs to assign the discrete label from the model to the continuous node created from the geometric SLAM algorithm.
- This combination produces a comprehensive 2D occupancy grid that is accurately labelled with objects from the environment.

Semantic SLAM System



This system will be running a object detection model and a graph SLAM algorithm on 4 cores and 8gb of RAM

Future Directions

The successful deployment of Semantic SLAM can lead to many significant implications in real-world situations. For example,
- Autonomous Navigation
  - Improving navigation in complex and dynamic environments [5].
- Long Term Autonomy
  - Enabling intelligent robot systems to adapt their surrondings and operate in various circumstances.
- Enhanced Virtual and Augmented Reality [7].
  - Improving the spatial mapping of different environments [4].
- Domestic Service Robotics
  - Upgrading existing consumer robots to execute more complex tasks involving the environment [6].

References

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